

.. Derive: OsteoLaus

.. Derive BMI ..

- . BMI derived from Weight and Height.
- . Total rows: 6114 | derived: 6113 | missing: 1
- . BMI range (non-missing): 15.1 - 45.5 kg/m²
- . Mean ± SD: 25.9 ± 4.7 kg/m²
- . Agreement with BMI_source (rows with both non-missing, n = 6112):
- . Difference < 0.01 (effectively identical): 6099 (99.8%)
- . Difference 0.01-0.5 (minor rounding): 13 (0.2%)
- . Difference >= 0.5 (meaningful mismatch): 0 (0%)
- . Median absolute difference: 0.002 kg/m²
- . Max absolute difference : 0.17 kg/m²

.. Deriving BMI category ..

- . derive_bmi_category: BMI categories derived.
- . Processed 6114 rows (1 missing BMI).
- Prevalence of Obesity: 18.5%
- Prevalence of Overweight + Obesity: 53.4%

.. Derive ALM Indices ..

.. Raw indices (native DXA platform)

- . 'ALM': 5635 derived, 479 missing
- . Range 9061.706 - 27682.402 |
- Mean ± SD: 17024.849 ± 2595.696
- . 'ALM_HT2': 5635 derived, 479 missing
- . Range 4.1 - 10.943 |
- Mean ± SD: 6.581 ± 0.86
- . 'ALM_BMI': 5635 derived, 479 missing
- . Range 0.379 - 1.054 |
- Mean ± SD: 0.668 ± 0.101
- . 'ALM_WT': 5635 derived, 479 missing
- . Range 0.178 - 0.381 |
- Mean ± SD: 0.258 ± 0.028

.. Harmonised indices (Lunar converted to Hologic scale)

- . 'ALM_harmonised': 5635 derived, 479 missing
- . Range 9061.706 - 27682.402 |
- Mean ± SD: 16487.142 ± 2625.595
- . 'ALM_HT2_harmonised': 5635 derived, 479 missing
- . Range 3.811 - 10.943 |
- Mean ± SD: 6.371 ± 0.874
- . 'ALM_BMI_harmonised': 5635 derived, 479 missing
- . Range 0.354 - 1.054 |
- Mean ± SD: 0.647 ± 0.102
- . 'ALM_WT_harmonised': 5635 derived, 479 missing
- . Range 0.168 - 0.381 |
- Mean ± SD: 0.249 ± 0.029
- . Sanity check passed: ALM_harmonised == ALM for all 2360 Hologic rows.
- . Agreement 'ALM' vs 'ALM_source' (n = 5635):
- . < 0.001: 5629 (99.9%)
- . 0.001 to 0.1: 0 (0%)
- . >= 0.1: 6 (0.1%)
- . Median |diff|: 0 | Max: 9.0683

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6 row(s) differ by >= 0.1 between derived and source.
. Agreement 'ALM_HT2' vs 'ALM_HT2_source' (n = 5635):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 6 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 0.0035
. Agreement 'ALM_BMI' vs 'ALM_BMI_source' (n = 5634):
. < 0.001: 5629 (99.9%)
. 0.001 to 0.1: 5 (0.1%)
. >= 0.1: 0 (0%)
. Median |diff|: 1e-04 | Max: 0.0052
. Agreement 'ALM_WT' vs 'ALM_WT_source' (n = 5635):
. < 0.001: 5635 (100%)
. 0.001 to 0.1: 0 (0%)
. >= 0.1: 0 (0%)
. Median |diff|: 0 | Max: 2e-04

.. Split ALM by DXA method ..

. Method-specific ALM columns created:
. Hologic and Lunar splits for ALM, ALM_HT2, ALM_BMI, ALM_WT

.. Derive Age ..

. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 6114 | derived: 6114 | missing: 0
. Age range (non-missing): 50.2 - 91.5 years
. Mean ± SD: 68.6 ± 7.9 years
. Agreement with Age_source (rows with both non-missing, n = 6114):
. Difference < 0.1 yr (effectively identical): 6066 (99.2%)
. Difference 0.1-1 yr (minor rounding/timing): 47 (0.8%)
. Difference >= 1 yr (meaningful mismatch) : 1 (0%)
. Median absolute difference: 0.028 years
. Max absolute difference : 4.328 years

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